Alessandro Armuzzi, Talat Bessissow, Timo Rath, Dong-Hoon Yang, Séverine Vermeire, Hiroshi Nakase, Raf Bisschops
Institutions: UZ Leuven, Belgium; Imelda vzw, Belgium; Katholieke Universiteit Leuven, Belgium; Griffith University - Gold Coast Campus, Australia; University College Cork, Ireland; Sapporo Ika Daigaku, Japan; Birmingham City University, United Kingdom; University College London, United Kingdom; Humanitas University, Italy; McGill University, Canada; Universitatsklinikum Erlangen, Germany; Seoul National University Hospital, Korea (the Republic of). Disclosure compliance: I understand. Participant disclosure: Pieter Sinonquel: NO financial relationship with a commercial interest; Peter Bossuyt: Advisory Committee or Review Panels: celltrion, dr falk benelux, galapagos, lilly, pentax, takeda, circ, globalport, psi cro, roche, tetrameros, janssen, abvvie; Grant/Research Support: abbvie, amgen, janssen, viatris, pfizer; Speaking and Teaching: dr falk benelux, epgs, galapagos, lilly, matiria prima, pentax, takeda, abbvie, janssen, bms, CAG, celltrion; Sneha John: Speaking and Teaching: Medtronic, 3D Matrix, Olympus Australia; Marietta Iacucci: Advisory Committee or Review Panels: Pfizer; Grant/Research Support: Olympus; Speaking and Teaching: PENTAX; Uday Shivaji: NO financial relationship with a commercial interest; Gert De Hertogh: Independent Contractor: Centocor J&J; Speaking and Teaching: Medtronic; Stephanie Van Aelst: NO financial relationship with a commercial interest; Sooraj Rajendran Pillai: NO financial relationship with a commercial interest; Zainab Abdawn: NO financial relationship with a commercial interest; Shintaro Sugita: NO financial relationship with a commercial interest; Sara McCartney: Speaking and Teaching: Abbvie, GSK; Alessandro Armuzzi: Consulting: Abbvie, Boehringer Ingelheim, Bristol-Myers Squibb, Eli-Lilly, Ferring, Galapagos, Janssen, Lionhealth, MSD, Nestlé, Pfizer, Protagonist Therapeutics, Takeda, Tillots Pharma; Speaking and Teaching: Abbvie, Bristol-Myers Squibb, Celltrion, Eli-Lilly, Ferring, Galapagos, Janssen, Lionhealth, MSD, Pfizer, Takeda; Talat Bessissow: Advisory Committee or Review Panels: Abbvie, Janssen, Bristol Myers Squibb, Eli Lilly, Pfizer, Sandoz, Takeda, Sanofi; Consulting: Alimentiv; Grant/Research Support: Abbvie, Janssen; Speaking and Teaching: Abbvie, Janssen, Amgen, Bristol Myers Squibb, Sandoz, Takeda, Ferring, Viatris, Fresenius Kabi; Timo Rath: NO financial relationship with a commercial interest; Dong-Hoon Yang: NO financial relationship with a commercial interest; Séverine Vermeire: Consulting: AbbVie, Abivax, AbolerIS Pharma, AgomAb, Alimentiv, Arena Pharmaceuticals, AstraZeneca, BMS, Boehringer Ingelheim, Celgene, Cytoki Pharma, Dr Falk Pharma, Ferring, Galapagos, Genentech-Roche, Gilead, GSK, Hospira, Imidomics, Janssen, J&J, Lilly, Materia Prima, Mestag Therapeutics, MiroBio, Morphic, MrMHealth, Mundipharma, MSD, Pfizer, Prodigest, Progenity, Prometheus, Robarts Clinical Trials, Surrozen, Takeda, Theravance, Tillots Pharma AG, VectivBio, Ventyx, Zealand Pharma.; Grant/Research Support: AbbVie, J&J, Pfizer, Takeda and Galapagos; Hiroshi Nakase: NO financial relationship with a commercial interest; Raf Bisschops: Advisory Committee or Review Panels: Norgine, Ipsen, Endostart; Consulting: Olympus, Pentax, Fujifilm, Abbvie, Ferring, Medtronic; Grant/Research Support: Pentax, Fujifilm, Medtronic, Norgine; Speaking and Teaching: Pentax, Medtronic, Norgine, Fujifilm.

Background: Red Density (RD) is an automated endoscopic tool that is developed for the objective evaluation of disease activity in ulcerative colitis (UC). Initial development was based on an algorithm including histological disease activity based on Robarts histological index (RHI). New histological scores for UC have been developed since.. **Aim**: To establish the correlation between RD and the Nancy histological index (NHI) and PICaSSO Histologic Remission Index (PHRI). Methods: Patients included in 4 centers in the ongoing PROCEED-UC study (NCT04408703) had assessment with RD in the rectum and sigmoid at baseline, w52 or early termination visit. Biopsies were taken according to protocol. Biopsies were scored for the Geboes score (GBS), RHI, NHI, PHRI in a blinded way after initial scoring convention training. Correlation was tested between mean RD per segment and the different histological indices based on Spearman correlation. Results: In total 96 patients from 4 centers were included representing 2634 RD images, 400 colonic segments with biopsies and corresponding RD score. Mean (± SEM) rectal RD score was 32.7 (± 4.54), 33.69 (± 5.25) and 64.86 (± 31.71) at baseline, w52 and ET respectively. Mean sigmoidal RD score was 38.13 (± 3.68), 42.82 (± 11.80) and 77.33 (± 24.65) at baseline, w52 and ET respectively. Analysis based on the segment with the highest mean RD score per patient showed significant correlation with NHI ($r=0.60$, $p<0.0001$), PHRI ($r=0.62$, $p<0.0001$). In the rectum the RD score at all time points correlated significant with NHI ($r=0,53$, $p<0.0001$) and PHRI ($r=0.63$, $p<0.0001$). Similar correlation was seen in the sigmoid for NHI ($r=0.51$, $p<0.0001$) and PHRI ($r=0.22$, p0.0108). A RD score of <57.5 (Likelihood Ratio (LR) 4.176; AUC 0.7820 (95CI 0.7047 - 0.8593), $p<0.0001$) was associated with histological remission based on NHI (<2) and a RD score of < 64.5 (LR 9.821; AUC 0.8133 (95%CI 0.7053 - 0.9212), $p<0.0001$) with histological remission based on PHRI (=0). The current dataset demonstrated to be stable in line with the previously established RD cut-off with GBS and RHI.[1]. Conclusion: In an independent cohort of patients with UC the correlation with the established and new histological indices is confirmed. This confirms the value of RD as objective endoscopic tool for the assessment of histological and endoscopic disease activity.

Sa1971

Artificial Intelligence Endoscopy

THE BOSTON ERCP DATASET: A VIDEO DATASET FOR ADVANCED ENDOSCOPY



Mark Geissler, Debesh Jha, Sami Elamin, Roya Dastjerdi, Fiona Kolbinger, Ulas Bagci, Moamen Gabr, Douglas Pleskow, Mandeep Sawhney, Tyler Berzin
Institutions: Beth Israel Deaconess Medical Center, USA; Harvard Medical School, USA; Northwestern University, USA; Purdue University, USA. Disclosure compliance: I understand. Participant disclosure: Mark Geissler: NO financial relationship with a commercial interest; Debesh Jha: NO financial relationship with a commercial interest; Sami Elamin: NO financial relationship with a commercial interest; Roya Dastjerdi: NO financial relationship with a commercial interest; Fiona Kolbinger: NO financial relationship with a commercial interest; Ulas Bagci: NO financial relationship with a commercial interest; Moamen Gabr: Consulting: Boston Scientific, Olympus, FujiFilm, AdaptivEndo; Douglas Pleskow: Consulting: Boston Scientific, Olympus, Fuji; Mandeep Sawhney: Speaking and Teaching: Immunovia Inc; Stock Shareholder: Allurion Inc; Tyler Berzin: Consulting: Medtronic, Wision AI, Boston Scientific, Magnetiq Eye, RSIP Vision.

Background: Despite the promise of artificial intelligence (AI) in GI endoscopy, computer-aided polyp detection (CADe) is the only category of algorithms to have achieved regulatory approval for GI endoscopy in the United States. Advancements in CADe and computer-aided diagnosis (CADx) for colon polyps are progressing rapidly, enabled by a plethora of colon polyp datasets, both proprietary and open-source. However, there is an unmet potential for AI algorithms applied to advanced endoscopic procedures. Comprehensive datasets for advanced endoscopy procedures such as endoscopic retrograde cholangiopancreatography (ERCP) are missing. Here, we describe the Boston ERCP Dataset, a first step towards a broader library of curated advanced endoscopy video datasets, with the goal of supporting AI innovation in advanced endoscopy. Material and Methods: ERCP videos were recorded from April to August 2023 at the Beth Israel Deaconess Medical Center. The recording was performed using Olympus endoscopy equipment (Olympus, Germany) and VIRGO (VIRGO, USA) recording devices. Videos were uploaded to the ideo annotation platform CVAT (CVAT,USA). During annotation labels and bounding boxes were provided. The initial labeling was performed by trained medical students and checked in a stepwise approach by gastroenterology fellows and subsequently attendings with extensive experience in the field. Additionally, corresponding fluoroscopy images, ERCP reports and pre-, intra- and post-procedural anesthesia reports were anonymized and added to the dataset. Results: The initial dataset comprises 208 ERCP procedure videos with an overall length of 76 hours accounting for over 7 million frames. Patient mean age was 65 years; 52% of participants were female. The most common indication for ERCP was obstruction of the bile duct (53%). The majority of cannulations were categorized as not difficult (88%). Labeling was focused on three general categories: procedural phase, instrument/device detection, and landmark recognition. Bounding boxes and segmentation masks were provided for instruments and anatomical landmarks (Figure 2). In addition, we provide anonymized reports for each procedure. These reports cover procedure reports, pre-, intra- and post-procedural anesthesia reports and if suitable histology reports. Additionally, we provided corresponding fluoroscopy images for each procedure. Conclusion: To our knowledge, this represents the first publicly available dataset focused specifically on advanced endoscopy. The curated videos and corresponding reports will serve as a resource for the development of AI tools for advanced endoscopy, to support procedural guidance, report generation and more.

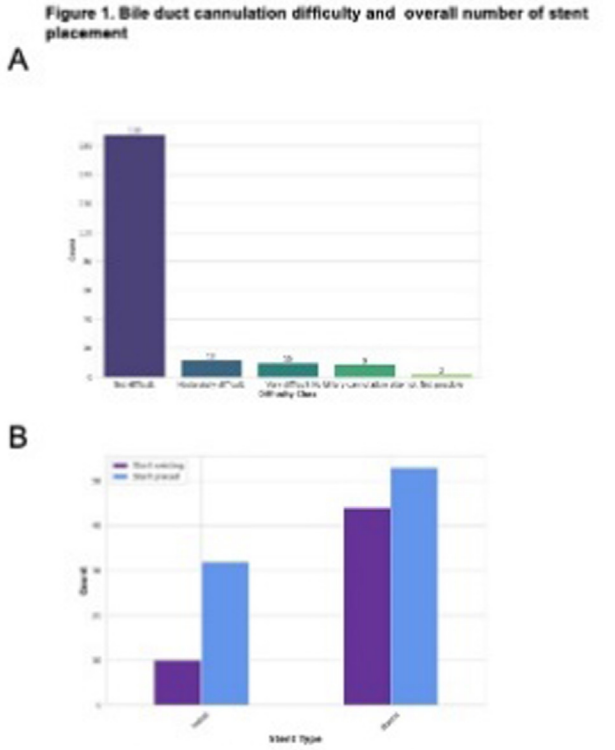


Figure 1. A) Difficulty of bile duct cannulation assessed by treating endoscopist B) Number and type of existing and placed bile duct stents

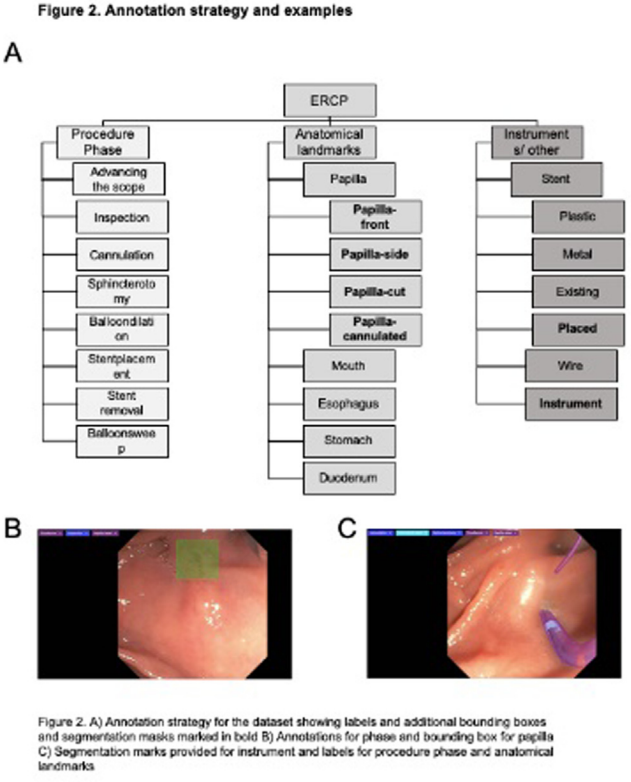


Figure 2. A) Annotation strategy for the dataset showing labels and additional bounding boxes and segmentation masks marked in bold B) Annotations for phase and bounding box for papilla C) Segmentation marks provided for instrument and labels for procedure phase and anatomical landmarks

Artificial Intelligence
Sunday, May 19, 2024
10:00 AM - 11:30 AM
Topic Forum

418

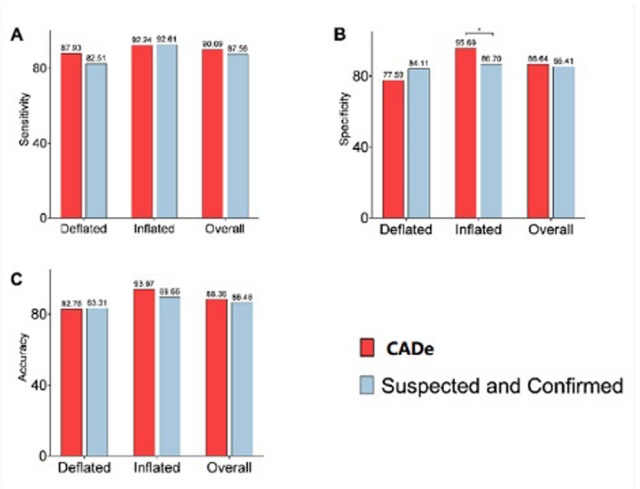
Ai in Endoscopy - New Technologies

ADVANCED COMPUTER-AIDED DETECTION SYSTEM EXHIBITS NO MORE FALSE POSITIVES THAN EXPERIENCED ENDOSCOPISTS IN COLONOSCOPY - AN IMAGE-BASED PROSPECTIVE COMPARATIVE STUDY

Pu Wang, Shan Lei

Institutions: Sichuan Academy of Medical Sciences & Sichuan Provincial People's Hospital, China. Disclosure compliance: I understand. Participant disclosure: Pu Wang: NO financial relationship with a commercial interest; Shan Lei: NO financial relationship with a commercial interest.

Background: Computer-aided detection (CADe) has demonstrated efficacy in polyp detection during colonoscopy, though the persistence of false positives poses a major hurdle to endoscopists' acceptance of this technology. We observed that endoscopists are equally prone to have "false positives", akin to those generated by CADe systems, which are subsequently dismissed upon close observation. The objective was to compare the false positives and overall diagnostic performance between CADe and experienced endoscopists through an image-based study. Methods: A prospective enrollment of 116 consecutive patients, undergoing colonoscopy and diagnosed with at least one pathologically confirmed small colon polyp, occurred between March 2020 and August 2020 at Sichuan Provincial People's Hospital. Images of each polyp were captured at a non-close distance following complete lumen inflation, followed by deflation to induce lumen wrinkling without lens movement and ensuring polyp visibility, then images were retaken. The same methodology was applied to the well-inflated and deflated lumen in polyp-free areas of the same colon segment. Randomly arranged images were labeled by a previously validated CADe system and a group of experienced endoscopists. Sensitivity, false positives, specificity, and accuracy were compared between the CADe system and endoscopists. Results: The study comprised 464 images from 116 patients with 116 polyps. The CADe system and endoscopists exhibited sensitivity, specificity, and accuracy of 90.9% vs. 87.56% (P=0.46), 86.64% vs. 85.41% (P=0.79), and 88.36% vs. 86.48% (P=0.43), respectively, across all labeled images. In the test set with images taken in fully inflated lumen, sensitivity, specificity, and accuracy for the CADe system and endoscopists were 92.24% vs. 92.61% (P=1.00), 95.69% vs. 86.70% (P=0.04), and 93.97% vs. 89.66% (P=0.13), respectively. In the test set with images taken in fully deflated lumen, sensitivity, specificity, and accuracy for the CADe system and endoscopists were 87.93% vs. 82.51% (P=0.35), 77.59% vs. 84.11% (P=0.24), and 82.76% vs. 83.31% (P=1.00).(figure 1, 2). Conclusion: The results indicate that the state-of-the-art CADe system exhibits no more false positives than experienced endoscopists while achieving higher sensitivity. Endoscopists' perception of increased false positives in CADe systems may stem from identical boxes appearing on both true polyps and non-polyp areas, while human endoscopists subjectively ignoring their own false suspicions due to the absence of visible boxes. Enhancements to CADe systems may involve aligning indications with human cognitive processes, pursuing the content of each box as meaningful as possible rather than chasing absolute zero false positives.



Performance of CADe and endoscopists